

DATABASES 2nd Course

Chapter I – Database theory – part 2

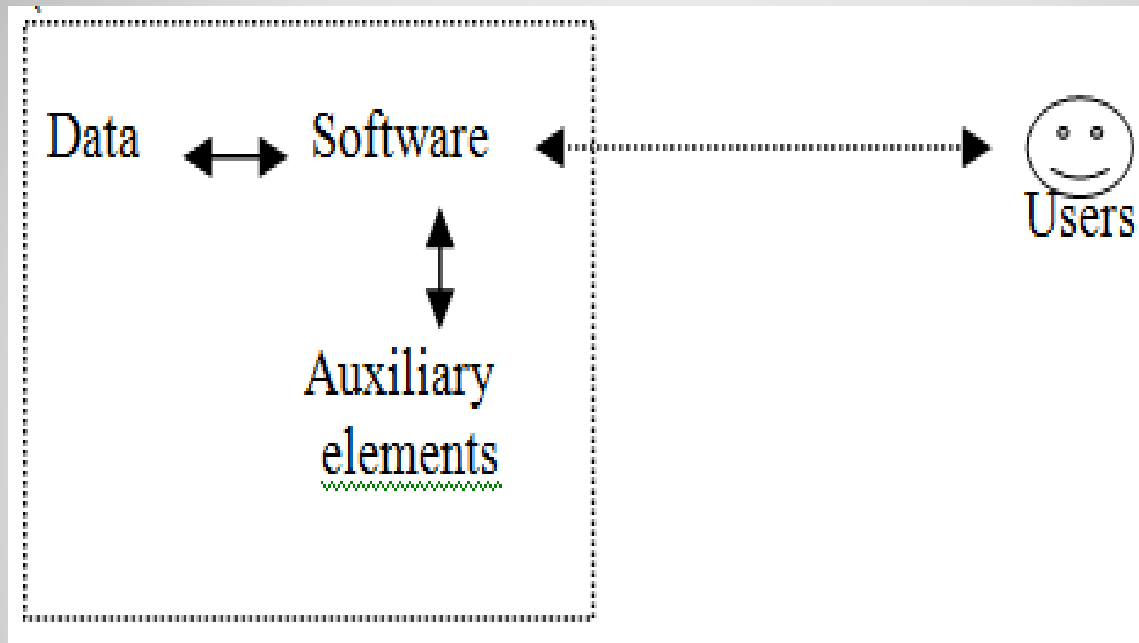
- **Definition**

- Database System (DBS) = a set of elements that contributes to the development and implementation of database applications.
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- The architecture of an DBS = graphical representation of the elements of a DBS and the links between them.

Database system

Two types of DBS architectures are used depending on what is graphically highlighted:

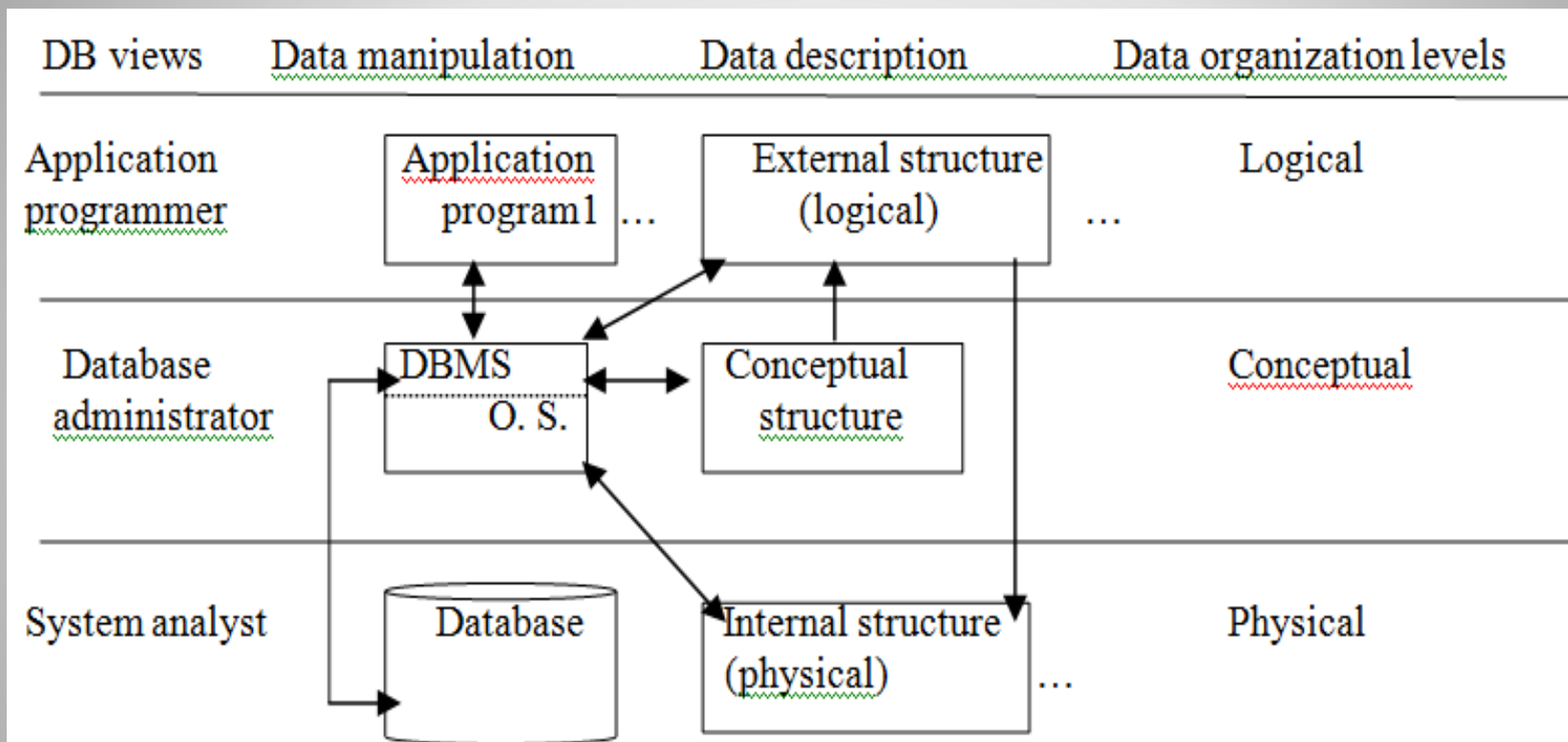
- **components architecture**
- **data organization tiered architecture.**



The components architecture gives an overview of the components of DBS and the interdependence between them.

DBS Components architecture

- 1. Data that is organized in a database according to a logical data model and contains:
 - Actual data collections;
 - Data dictionary;
 - Related files.
- 2. Software that is necessary for developing and implementing the database and contains:
 - database management system (DBMS);
 - application programs that are mostly written in DBMS's languages.
- 3. Auxiliary elements, which are components that contribute to the development and functioning of the entire DBS:
 - set of automatic procedures and manuals;
 - legal and administrative regulations;
 - hardware resources that are used;
 - involved staff that is split in user groups.



The tiered architecture structures a DBS on three levels and gives an insight of its organizing and functioning mode.

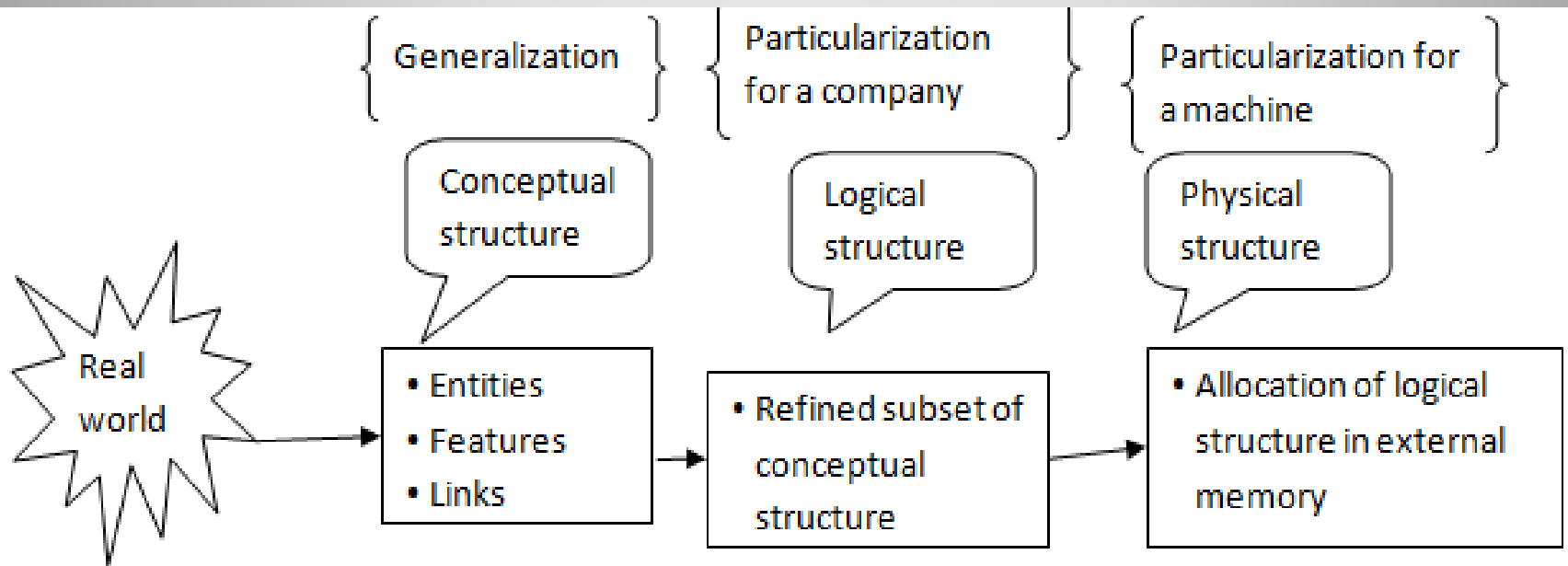
DBS Tiered architecture

The tiered architecture gives the levels of organizing (representing) data in the database:

- 1. Conceptual level is given by the database administrator's view over data:
 - the administrator performs the conceptual structure of DB, possibly with tools provided by DBMS;
 - the conceptual structure is obtained using a specific data model for DB and an appropriate design technique;
 - the conceptual structure is a representation of reality, within the system, that a DB has transcribed it;
 - the administrator's view over the database is independent of the applications that will be developed (logical independence);
 - the result of the conceptual level is the conceptual schema;
 - the schema's development corresponds to a modeling activity because it is a translation in abstract terms of real-world entities.

- 2. Logical level is given by the programmer's view over the DB:
- the programmer develops application programs for describing and manipulating data, that are written in a DBMS;
- the programs implement the external (logical) structure of data;
- the external structure comes from the conceptual structure;
- the external structure is a programmer's view over the DB for a certain application;
- the programmer's view is independent of technical information support (physical independence);
- the result of a logic level is an external schema, as part of the conceptual schema, implemented using a DBMS.

- 3. Physical level is given by the system analyst's view over data:
 - the system analyst develops the internal (physical) structure;
 - the internal structure comes from external one, using allocation techniques on physical support;
 - the internal structure corresponds to data description on physical information support;
 - at physical level, the result is the internal (physical) schema, which is defined in terms of files and records;
 - internal schema implementation is done using the file management system from DBMS and / or OS, through physical management of peripherals.



• Definition

DBMS = a set of programs (software) that interfaces between a database and its users and provides the following activities:

- defining the database structure;
- loading / validation of data onto the database;
- data access (retrieve, update);
- DB maintenance (retrieval, recovery, conversion etc.);
- data protection.

DBMS

Functions of a DBMS

- In order to achieve its goals, any DBMS has four mandatory (fundamental) functions that are of a high degree of generality.
- 1. Defining data - allows describing and defining the data structure of a DB (entities, their characteristics, the links between data, data integrity) using a DDL (Data Description Language) for conceptual, logical, and physical level.

The result of the defining function is the DB schema that is stored in internal code (DB dictionary).

- 2. Data manipulation – allows loading data in the database, retrieving data (from very simple to complex criteria), modifying data (inserting, updating, deleting), with a DML (data manipulation language).
- 3. Using data - provides the interfaces (generators, browsers, etc.) needed for the communication of all users with the database.
- 4. Data administration - allows the maintenance and management of DB.

- DB administrator = one or more people with experience in analysis and design, dealing with organizing and maintaining the database.
- Types of DB administrators regarding their activities:
 - *global administrator* achieves:
 - DB (conceptual) schema, resulted from the applications' requirements;
 - organizing data at the logical level (collaboration).
 - *application administrator* performs:
 - external (logical) schema for an application, resulted from processing requirements;
 - organizing data at physically level (collaboration).
 - *database administrator* makes:
 - internal (physical) schema of data;
 - reorganization of the database;
 - managing the DB functioning.

DB
administration

- Database administrator tasks are structured like the DB developing steps.
- **- for database analysis and design:**
 - defines the objectives of DB;
 - collaborates in formulating application requirements;
 - defines DB dictionary (schema, integrity constraints, etc.);
 - cooperates for the internal and external schema;
 - designs data protection.

- **- for DB implementation:**

- develops documentation;
- defines rules for the implementation and putting to use of DB;
- ensures DB loading from different data sources.

- **- for DB exploitation:**

- monitoring the data access;
- ensures data protection;
- maintains the DB within the designed parameters.

Tools available for the database administrator to perform tasks:

- tools provided by the DBMS for: DB reorganization, DB restoration, statistical analysis, managing the data dictionary, data protection.
- specific tools created by administrators and programmers.